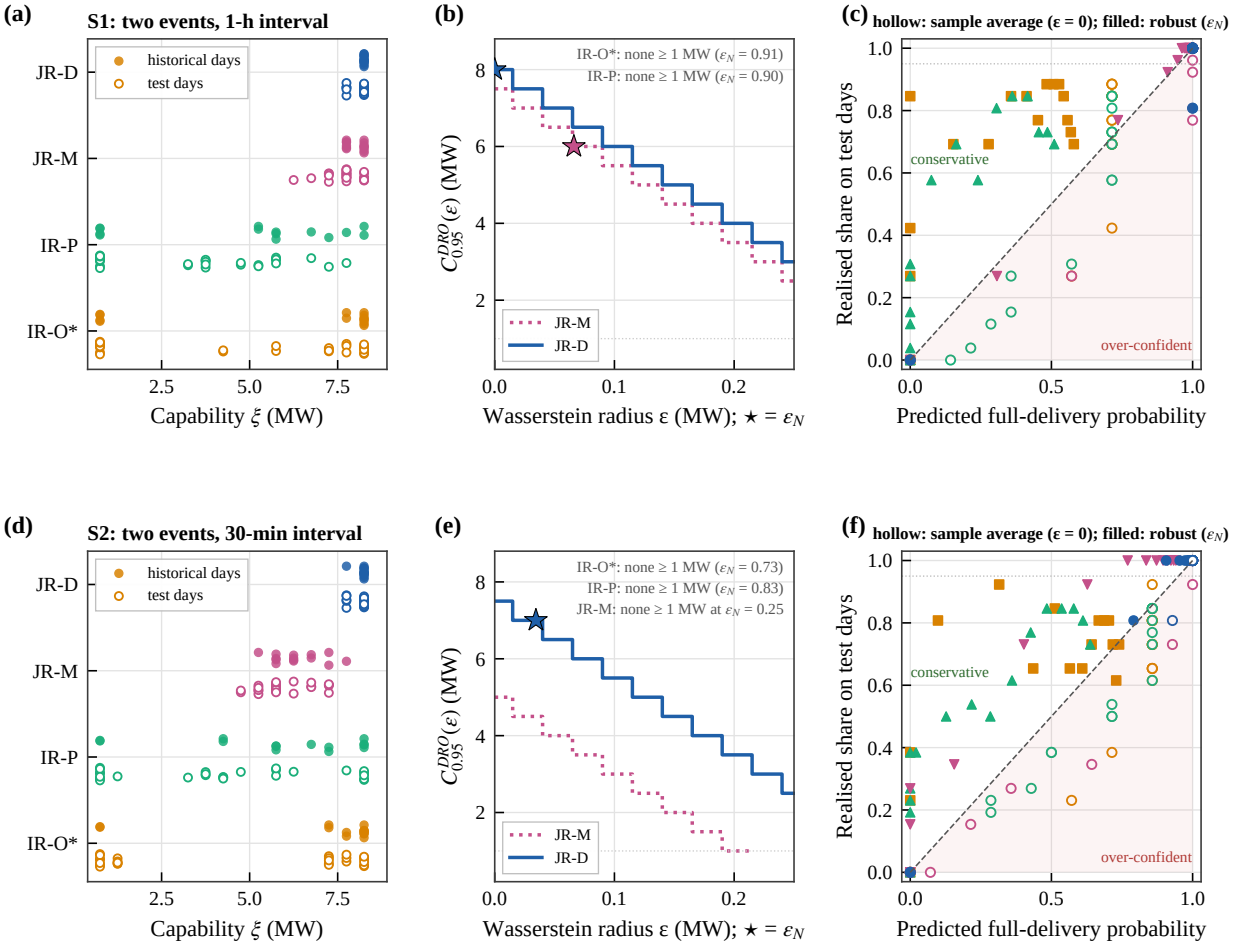
(a)
S1: two events, 1-h interval
historical days
test days
JR-D
JR-M
IR-P
IR-O*
2.5
5.0
7.5
Capability ξ (MW)
(b)
IR-O*: none ≥ 1 MW (ε_N = 0.91)
IR-P: none ≥ 1 MW (ε_N = 0.90)
$C^{DRO}_{0.95}(\varepsilon)$ (MW)
JR-M
JR-D
0.0
0.1
0.2
Wasserstein radius ε (MW); ★ = ε_N
(c)
hollow: sample average (ε = 0); filled: robust (ε_N)
Realised share on test days
conservative
over-confident
0.0
0.5
1.0
Predicted full-delivery probability
(d)
S2: two events, 30-min interval
historical days
test days
JR-D
JR-M
IR-P
IR-O*
2.5
5.0
7.5
Capability ξ (MW)
(e)
IR-O*: none ≥ 1 MW (ε_N = 0.73)
IR-P: none ≥ 1 MW (ε_N = 0.83)
JR-M: none ≥ 1 MW at ε_N = 0.25
$C^{DRO}_{0.95}(\varepsilon)$ (MW)
JR-M
JR-D
0.0
0.1
0.2
Wasserstein radius ε (MW); ★ = ε_N
(f)
hollow: sample average (ε = 0); filled: robust (ε_N)
Realised share on test days
conservative
over-confident
0.0
0.5
1.0
Predicted full-delivery probability